Project Initialization and Planning Phase

Date	30 September 2026
Team ID	SWTID-2026-2032
Project Name	Learnmate – AI StudyBuddy
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

A customer problem statement helps us see the problem from the customer's point of view. It keeps the team focused on what matters, builds empathy for the people we are designing for, and guides us toward the ideal solution for their challenges.

Learnmate (AI StudyBuddy) is a web-based platform where students upload their study material (.txt, .md, .pdf) and receive AI-generated summaries, flashcards, quizzes, and personalized study plans, powered by Google Gemini.

Template

I am	I'm trying to	But	Because	Which makes me feel
Customer / persona	Goal they want to achieve	Barrier that stops them	Root cause of the barrier	Emotional impact

Example

I am	I'm trying to	But	Because	Which makes me feel
A student	Revise a long chapter before an exam	It is too long to read and condense in time	I have no quick way to extract key points	Overwhelmed and anxious

Problem Statements

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A college student with large PDFs and lecture notes	Quickly understand and revise my study material	Reading and summarizing everything manually takes hours	I have limited time and no tool to condense the content	Overwhelmed and stressed
PS-2	A student preparing for exams	Test how well I understand a topic	Creating flashcards and quiz questions by hand is slow and tedious	I don't know which concepts are most important to practice	Unprepared and unsure of my progress
PS-3	A self-learner with a lot of material to cover	Follow a clear day-by-day study schedule	I don't know how to split the material into manageable parts	I have no structured plan tailored to my content	Lost and unmotivated
PS-4	An administrator of the platform	Manage users and uploaded study materials	There is no secure, central way to oversee accounts and content	Access control and moderation are not built into simple study tools	Concerned about security and misuse

How Learnmate Addresses Each Problem

PS	Solution Feature	Implementation
PS-1	AI-generated summaries	Upload material (max 5 MB); Gemini returns a concise summary
PS-2	Flashcards and quizzes	Auto-generated Q&A flashcards and multiple-choice quizzes stored per material
PS-3	Personalized study plan	AI builds a study plan from the uploaded content
PS-4	Role-based access control	JWT authentication with student and admin roles; admin-only routes

Tech Stack

Layer	Technology
Backend	Node.js, Express.js
Database	MongoDB with Mongoose
Authentication	JWT, bcryptjs (password hashing)
File Uploads	Multer (.txt, .md, .pdf; 5 MB limit)
AI Engine	Google Gemini API (retry with backoff on 429/503 errors)
API Testing	Postman

API Routes

Route	Purpose	Access
/api/auth	Register and login	Public
/api/materials	Upload material; generate summary, flashcards, quiz, study plan	Authenticated users
/api/admin	Manage users and materials	Admin only

Project Structure

```
Learnmate/
`-- Code Files/
    |-- src/
    |   |-- controllers/  adminController.js, authController.js, materialController.js
    |   |-- middleware/   auth.js, upload.js
    |   |-- models/       Material.js, User.js
    |   |-- routes/       admin.js, auth.js, materials.js
    |   `-- utils/        db.js, gemini.js, tokens.js
    |-- uploads/
    |-- index.js
    |-- package.json
    `-- README.md
```

Setup

```
git clone <your-repo-url>
cd "Learnmate/Code Files"
npm install

# .env
MONGO_URI=your_mongodb_connection_string
JWT_ACCESS_SECRET=your_jwt_secret
GEMINI_API_KEY=your_gemini_api_key
CLIENT_URL=http://localhost:3000
PORT=5000

node index.js
```

Note: Never commit your .env file. Add .env, node_modules/, and uploads/ to .gitignore.